

# Samuel (Seongwoo) Park

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## EDUCATION

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Virginia Polytechnic Institute and State University

**B.S. in Electrical Engineering — Controls, Robotics, and Autonomy**

**M.S. in Computer Engineering — Software and Machine Intelligence**

- **Coursework:** Digital Image Processing, Machine Learning, Computer Vision, Reinforcement Learning, Computer Vision in Smart Manufacturing, Application of Machine Learning, Real-Time Systems, Principles of Robotics Systems

## WORK EXPERIENCE

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**RoboSeasy (Startup) – Robotics Engineer** (July 2025 - August 2025)

- Collected 200 demonstration and trained an Action Chunking Transformer (ACT) model on 6 DoF robotic Leader and Follower arm and automated the manipulation tasks achieving 97% success rate.
- -Executed a real-time simulated environment in NVIDIA Isaac Sim integrating a leader-follower arm setup where the leader arm controlled the simulated follower arm.

**Adroit – Robotics Engineer (August 2025 - August 2025)n (RoboSeasy)** (July 2025 - August 2025)

- Implemented a teleoperation pipeline linking the 6-DOF robot and 7-DOF robot via the HuggingFace Lerobot framework.

**Strayos -Computer Vision Engineer** (May 2026 - Present)

- Implementing object segmentation using ZED stereo camera attached to a loader to classify the oversized rock while diggin and alert the driver
- Using the ZED stereo camera map the mining field by implementing Visual SLAM to map quarry environments for visualization

## PROJECT

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**Undergraduate Research** (May 2023 - August 2023)

*Analyzing Construction Pollution Using Object Detection*

- Trained a YOLO-based model in PyTorch by labeling 500+ images of construction vehicles, achieving 92.5% mAP, 93.4% precision, and 92.0% recall
- Collected air quality data using Enviro+ on a Raspberry Pi and Python scripts; analysis indicated a positive correlation between vehicle presence and elevated pollution levels.

**AI-Agent Ice Cream Dispensing Robot** (October 2024 - November 2024)

- Built an object-detection pipeline to detect and localize the ice cream cup and verify its presence at the counter, enabling autonomous grasping and dispensing.
- Designed and implemented robot motion control and task-level state logic for an AI-agent–operated ice cream dispensing robot on 6-DOF robotic arm.

**Human Following AI Caddie** (November 2024 - December 2024)

- Utilized LiDAR sensor to integrate object avoidance algorithm for safe navigation of a 4 wheeled robot while following human.

**Autonomous Excavator Prototype** (October 2025 – May 2026)

- Prototyped a vision-guided excavator by integrating servo motors, Jetson Orin nano, stereo camera, linear actuators to automate limestone excavation.
- Designed and fabricated rubber track using Fusion 360 and tested both PLA, PETG and Urethane materials on limestones.

**Virginia Tech Autodrive Team** (September 2025 – May 2026)

- Implemented YOLOv11 integrated with Optical Character Recognition for real-time detection and classification of traffic signs.

**Microscopic Defect Classification using U-net and Attention U-net** (January 2026 – May 2026)

- Implemented and compared Attention U-Net and U-net in Pytorch for defect segmentation on the NanoTWICE SEM nanofiber dataset, building a full training pipeline with custom dataset.

## SKILLS

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- Robotics & Hardware: LiDAR, ROS2, Isaac Sim, MATLAB, Raspberry Pi, Jetson Orin Nano, RealSense camera, ZED camera
- Programming & Deep Learning: Python, C, PyTorch, YOLO